

ANNEXURE II

TAMILNADU STATE COUNCIL FOR SCIENCE AND TECHNOLOGY
DOTE Campus, Chennai – 600 025

Innovation and Product Development Scheme (IPD)

INNOVATION PRODUCT/PROJECT PROGRESS REPORT (For Every Three Months)

IMPORTANT POINTS:

- i. 2 copies of the report are required.
- ii. Report should be sent even if Innovation Product/Project has not become fully operational.
Please write 'Nil' against items where there is nothing significant to report or if these items are not relevant.
- iii. Report should cover the work done during the period of every three months.
- iv. Timely submission of report is essential to facilitate release of funds.
- v. Report should be in the format given below with cover page followed by report under the sections indicated below.
- vi. The Report should be discussed and finalised by the Innovation Product/Project team before dispatch to the Council.



தமிழ்நாடு அறிவியல் தொழில்நுட்ப மாநில மன்றம்
TAMIL NADU STATE COUNCIL FOR SCIENCE AND TECHNOLOGY

(Established by Government of Tamil Nadu)
Directorate of Technical Education Campus, Chennai - 600 025.

Phone : 044 - 2230 1428
: 044 - 2230 1552

Web : www.tanscst.tn.gov.in
E-mail : ms.tanscst@nic.in / tncouncil18@gmail.com

Prof. Dr. S. VINCENT, D.Sc., Ph.D., MNASc., FNABS., FRSB (Lon)

Member Secretary

Letter No: TNSCST/ IPD-076 /2025-26

19.11.2025

To

Ragavi Natarajan,
Department of Biomedical Engineering,
E.G.S. Pillay Engineering College,
Nagapattinam – 611002

Sir,

Subject: Innovation and Product Development (IPD) Project on “Enhancing existing Knee Cap with added foam cushion support for PCL (Posterior Cruciate Ligament)”- Approval – Intimation – Regarding.

Ref: Your application dated: 20.03.2025

I am happy to inform you that TNSCST has approved your above IPD Proposal for funding. The approved budget for the project entitled “Enhancing existing Knee Cap with added foam cushion support for PCL (Posterior Cruciate Ligament)” is Rs. 85095/- (Rupees Eighty-five thousand ninety-five only) for one year. A copy of the terms and conditions and budget breakup is enclosed herewith for your reference. Kindly acknowledge your acceptance on or before 28.11.2025 (through mail) for the above budget details and terms and conditions.

You have to submit a declaration stating that the above project has not received any funds from any other funding agency. All communication should be sent through the proper channel through mail.

Kindly provide your Institutional Bank account details such as Name of the A/C, Account Number, Account type (SB or CA), IFSC Code, Name of the Bank and Branch. (with attested copy of bank details to mail)

With best wishes,

Yours faithfully,


Member Secretary

Encl: 1. Terms and Conditions with annexure (I to IV)
2. Budget Breakup

Copy to:

The Principal
E.G.S. Pillay Engineering College,
Nagapattinam – 611002

Dr.R.Venkatesan,
Associate Professor,
Department of Biomedical Engineering,
E.G.S. Pillay Engineering College,
Nagapattinam – 611002.

ENHANCING EXISTING KNEE CAP WITH ADDED FOAM CUSHION SUPPORT FOR PCL (POSTERIOR CRUCIATE LIGAMENT)

No: IPD 076

Date: 10-03-2026

Date of Start: 12-12-2025 Total Cost of the Innovation Product/Project: Rs. 85,095/-

Report of the period: From: 12-12-2025 to 12-03-2026

Name of the Principal Investigator: Ms. Ragavi Natarajan

Name of the Institution: E.G.S. Pillay Engineering College

A.	PRELIMINARY:		
1.	Title of the Innovation Product/Project	:	Enhancing Existing Knee Cap with Added Foam Cushion Support for PCL (Posterior Cruciate Ligament)
2.	a) Name & Address of Innovation Product/Project Implementing agency	:	Department of Biomedical Engineering, E.G.S Pillay Engineering College,, Old Nagore Road, Nagapattinam – 611002
	b) Name of Principal Investigator	:	Ms. Ragavi Natarajan
3.	Name of Council Innovation Product/Project Officer	:	
4.	a) Duration of the Innovation Product/Project (as approved)	:	1 year
	b) Date of actual commencement:	:	12.12.2025
5.	Objectives	:	1. To design a concentric foam-supported kneecap brace for improved knee stabilization. 2. To develop and fabricate a prototype knee brace for chronic knee pain and PCL-related instability. 3. To evaluate the brace for pain reduction, stability, comfort, and posterior knee support. 4. To analyze the effectiveness of the design through preliminary user evaluation. 5. To disseminate the research findings.
6.	Target date fixed for completion of Innovation Product/Project)		December, 2026
7.	Approved financial outlay	:	Rs 85095.00
8.	Approved breakup of grants	:	
	Cost required for (Equipment, Fabrication, Testing, etc.)	:	75095
	Consumables (Contingency/overhead, miscellaneous and travel)	:	10000
	Others (Please list)		Nil
9.	Date of approval of Innovation Product/Project	:	19.11.2025
10.	Grants received so far (Date & Amount)	:	12.12.2025 Rs. 85095.00
11.	Summary of Progress reported in the previous reports	:	NIL (First progress report)
B.	PROGRESS (A) PHYSICAL		

	Work Completed so far and achievement as regards	:	1. Design and Concept Development ○ Conceptualized and designed a concentric foam-supported kneecap brace intended to improve stability and reduce pain in patients with chronic knee pain and posterior cruciate ligament (PCL)–related instability. ○ Developed the structural design focusing on biomechanical support, pressure distribution, and improved comfort during prolonged usage. 2. Prototype Development ○ Fabricated the initial prototype incorporating layered foam support and concentric stabilization around the patellar region. 3. Preliminary Evaluation ○ Conducted preliminary user evaluation to assess parameters such as: ▪ Pain reduction ▪ Stability improvement ▪ Comfort during usage ▪ Posterior knee support ○ Observed notable improvement in user-reported outcomes, indicating potential clinical applicability. 4. Research Dissemination ○ Presented the research findings titled at the International Conference on Health and Disease Management 2026 (ICHDM-2026). 5. Recognition / Award ○ Received the Best Oral Presentation Prize at the conference organized by the Department of Chemistry & Biosciences, SASTRA Deemed University – Srinivasa Ramanujan Centre, Kumbakonam.
	a. each of objectives listed at A (5):	:	Objective 1: Design of Knee Brace A concentric foam-supported kneecap brace was designed to provide improved patellar stabilization and posterior knee support for patients with chronic knee pain and PCL-related instability. Objective 2: Prototype Development A functional prototype of the knee brace was fabricated using layered foam and supportive structural components to enhance comfort, flexibility, and stability. Objective 3: Preliminary Evaluation Initial user evaluation was conducted to assess pain reduction, stability, comfort, and posterior support, showing positive improvements. Objective 4: Dissemination of Research The research findings were presented at the International Conference on Health and Disease Management 2026 organized by SASTRA Deemed

			University, where the work received the Best Oral Presentation Prize.
	b. each phase indicated in item A (6)	:	Phase 1: Design of the concentric foam-supported kneecap brace. Phase 2: Fabrication of the prototype using suitable foam and supportive materials. Phase 3: Preliminary evaluation for pain reduction, stability, and comfort. Phase 4: Development of the brace using appropriate biomaterials. Phase 5: Presentation of research findings

iii. Procurement / usage of equipment :

Sl.No.	Name of the equipment acquired	Make & Model	Procurement dated	Utilisation rate (100% 75% 50%)	Remarks Regarding service maintenance breakdown etc.
NIL					

iv. impediments / difficulties

v. Results reported during the year

a. Papers published / communicated in Scientific Journals

Sl.No.	Name of the Authors acquired	Title of the Paper	Name of the Journal	Vol / Page	Year
1.	Ragavi Natarajan, Venkatesan.R	Design, Development, and Preliminary Evaluation of Concentric Foam-Supported Kneecap Brace for Chronic Knee Pain and Posterior Cruciate Ligament-Related Instability.	Asian Journal of Pharmaceutical and Clinical Research (AJPCR)	Submitted, yet to be published.	2026

b. Participation in conferences / Seminars

Sl.No.	Name of the Conference With dates and Venue	Title of the paper if Presented	Name of the team Members who are attended
1.	International Conference on Health and Disease Management (ICHDM – 2026) on 26.02.2026 and 27.02.2026 at Sastra Deemed to be University, Srinivasa Ramanujan Centre, Kumbakonam	Design, Development, and Preliminary Evaluation of Concentric Foam-Supported Kneecap Brace for Chronic Knee Pain and Posterior Cruciate Ligament-Related Instability.	Ragavi Natarajan Venkatesan

(b) Financials

(i) Amount spend under each Subhead

Item	Amount
Conference Registration	₹750
Total	₹750

(ii) Balance available as on date of report

Total Grant = ₹85,095

Spent = ₹750

Balance = ₹84,345

(iii) Remarks, if any - NIL

(c) General

(i) Is any assistance required from the Council (Other than financial)

(ii) Visits of Scientists /Industry Experts and interaction with other agencies

(a) Give names of the Scientists/Industry Experts from other institutions whom you may have invited in connection with the Innovation Product/Project

(b) In case the results of the Innovation Product/Project are of likely use to any agency indicate progress towards promoting interaction with them.

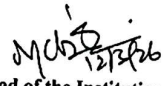
iv. Self assessment of progress.

(a) Does the Innovation Product/Project team feel that the Innovation Product/Project can be completed within the approved duration.

Yes, the project can be completed within the stipulated duration.

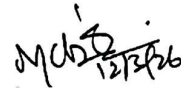
(b) If not, steps to be taken for speeding up the work may be suggested.

v. Date (s) on which the report was discussed by the Innovation Product/Project team:

Project team & Name	Signature	Date
Principal Investigator Ms Ragavi Natarajan		12-03-2026
Principal Co-Investigator Dr. R. Venkatesan	 Dr. R. VENKATESAN, M.Tech., Ph.D., Associate Professor Department of Biomedical Engineering E.G.S. Pillay Engineering College Nagapattinam - 611 002.	12-03-2026
Members of the E.G.S Pillay Engineering Innovation Product College	 Head of the Institution Dr. M. CHINNADURAI, M.E., Ph.D., PRINCIPAL E.G.S. Pillay Engineering College Nagore (Po), Nagapattinam - 611 002.	12-03-2026



Principal Investigator



Head of the Institution
Dr. M. CHINNADURAI, M.E., Ph.D.,
PRINCIPAL
E.G.S. Pillay Engineering College
Nagore (Po), Nagapattinam - 611 002.



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THANJAVUR | KUMBAKONAM | CHENNAI
SRINIVASA RAMANUJAN CENTRE, KUMBAKONAM



DEPARTMENT OF
CHEMISTRY & BIOSCIENCES



CERTIFICATE OF APPRECIATION

This certificate recognizes

Ms. Ragavi Natarajan

from

SASTRA Deemed University, Thanjavur

for her exemplary achievement as the recipient of the
Best Oral Presentation Prize at the International
Conference on Health and Disease Management
(ICHDM 2026), held on February 26 and 27, 2026. Her
outstanding scholarship, excellence in presentation,
and significant contributions to the advancement of
knowledge are highly commended and deeply
appreciated.

Dr. R. Sivakumar
Convener

Dr. B. Santhi
Dean



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भारत का नवोद्यम
The Innovation Engine of India

DEPARTMENT OF
CHEMISTRY & BIOSCIENCES



WITH MAXIMUM BONDING

CERTIFICATE OF APPRECIATION

This certifies that

Ms. Ragavi Natarajan

from

SASTRA Deemed University, Thanjavur

delivered an oral presentation of her research findings at the International Conference on Health and Disease Management (ICHDM 2026), held on February 26 and 27, 2026.

We extend our sincere appreciation for her enthusiasm, dedication, and meaningful engagement throughout the event.

R. Sivakumar
Dr. R. Sivakumar
Convener

Dr. B. Santhi
Dr. B. Santhi
Dean



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THANJAVUR KUMBAKONAM CHENNAI



CSIR, Govt. of India Sponsored

**International Conference on Health
and Disease management (ICHDM-2026)**

February 26th and 27th, 2026

Organised by

**DEPARTMENT OF CHEMISTRY
& BIOSCIENCES**

SRINIVASA RAMANUJAN CENTRE
KUMBAKONAM

Appreciation Award